PART II · WHEN, AND FOR HOW LONG

Time

Music never stands still. Underneath runs a steady **pulse**, packed into **bars**, and every note and rest gets its fair share. We start with the big beat and zoom in to a single note.

Pulse and bar the steady beat, in groups of 2, 3 or 4 › **Note values** how long each sound and silence lasts › **Longer notes** dots and ties › **Time signature** how many beats, and which note is one beat › **Rhythm** long and short, on and off the beat

IN THIS PART

Pulse, tempo and the bar 34
Notes and rests 36
How long is a note? 39
Time signatures 41
Compound and irregular time 43
Rhythm: groups and upbeats 45
Rhythm: off the beat 47
Tips: when the rhythm slips 49

PART II · TIME · 1

Pulse, tempo and the bar

Put your hand on your heart: that steady beat is a **pulse**, and almost every piece has one. Its speed is the **tempo**. Beats travel in groups called **bars**, and beat 1 is the strongest.

THE SECRET

The beat never stops, not even in a rest. Feel it before you play.

Tempo

Tempo is measured in beats per minute. A **metronome mark** at the top of the music shows it: ♩ **= 60** means 60 crotchet beats a minute, one beat a second. ♩ **= 120** is twice as fast. Tempo words such as *Andante* and *Allegro* come with Expression, later in the book.

Strong and weak beats

BEATS	PATTERN	FEELS LIKE
2	**Strong** weak	a march
3	**Strong** weak weak	a waltz
4	**Strong** weak **medium** weak	most songs

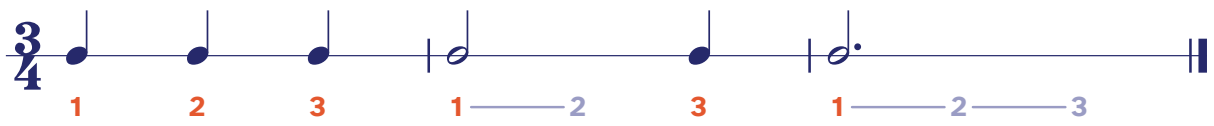
Bars and bar lines

SIGN	NAME	MEANING
𝄀	bar line	divides the music into bars
𝄁	double bar line	marks the end of a section, or a change of key, time signature or tempo
𝄂	final bar line	marks the end of the piece
𝄆 𝄇	repeat signs	play the passage between them again; if there is no start sign, go back to the beginning

The beat in tunes you know

TUNE	BEATS IN A BAR	PATTERN
Twinkle, Twinkle, Little Star	4	**Strong** weak **medium** weak
Happy Birthday	3	**Strong** weak weak
The Blue Danube (a waltz)	3	**Strong** weak weak
Radetzky March	2	**Strong** weak

Counting



Three bars of 3/4. Every beat is counted, even inside a long note: the red number is where you play, the grey numbers are the beats you hold.

PART II · TIME · 1 · CONTINUED

Count aloud at one speed, a little louder on beat 1. With a metronome, start slower than you think and speed up only when the passage is steady.

EXAMPLE 1

Is this a march or a waltz?



1. **Tap a steady beat** as you clap or listen.
2. **Find the strong one:** the red beats feel heavier.
3. **Count the gap** between strong beats: 1 2 3, 1 2 3.

Groups of three: a **waltz**, in 3/4. Groups of two would be a march.

EXAMPLE 2

Feel the beat before you play
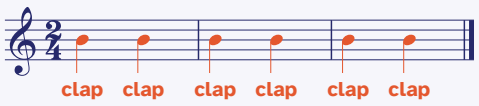


1. **Read the mark:** ♩ **= 96**, 96 crotchet beats a minute.
2. **Listen first:** set the metronome and let one full bar go by, counting “1 2 3” in your head.
3. **Step in** on the next beat 1, the way you step onto a moving escalator.

The first note lands **on the beat**, not a guess.

YOU KNOW THIS

Radetzky March (Johann Strauss I)



At the New Year concert the audience claps along on **every beat**. That steady clap is the pulse; the orchestra’s tune moves freely over it.

Try it. Put on any march and clap only the beat, not the tune.

THINK OF IT The beat is the stage and the notes are the actors: the stage is there before anyone walks on. So step in the way you step onto an escalator or take a plate off a sushi train: watch it move for a bar, then join in.

TRAP

- × Speeding up in the easy bars and slowing down in the hard ones: the pulse stays the same all the way through.
- × Counting only the notes you play: the beat goes on through long notes and rests.

SEE ALSO

Time signatures p. 41 · Tempo words: Expression (later)

PRACTISE IT

Workbook Level 0 · Pulse · Online: Metronome

PART II · TIME · 2

Notes and rests

You can see how long a note lasts from its shape: each plain note is **half** the one above it. And every note has a twin **rest**, a silence you count just as carefully.

THE SECRET

Each note is half the one above it, and every note has a rest twin.

NOTE	REST	NAME	ALSO CALLED	BEATS*	FIRST MET
		**breve**	double whole note	**8**	Level 4
		**dotted semibreve**	dotted whole note	**6**	*Extra*
		**semibreve**	whole note	**4**	Level 0
		**dotted minim**	dotted half note	**3**	Level 0
		**minim**	half note	**2**	Level 0
		**dotted crotchet**	dotted quarter note	**1½**	Level ½
		**crotchet**	quarter note	**1**	Level 0
		**dotted quaver**	dotted eighth note	**¾**	Level 3
		**quaver**	eighth note	**½**	Level 0
		**semiquaver**	sixteenth note	**¼**	Level 2
		**demisemiquaver**	32nd note	**⅛**	*Extra*
		**hemidemisemiquaver**	64th note	**1/16**	*Extra*

* With the crotchet as one beat, as in 2/4, 3/4 and 4/4. *Extra*: beyond Level 8.

PART II · TIME · 2 · CONTINUED

Why whole, half and quarter?

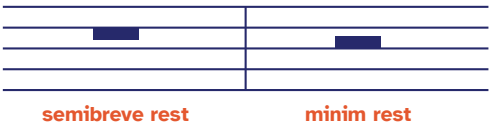
Think of each beat as a pizza. A bar of 4/4 is four pizzas. The note that fills the whole bar is the **whole note**; the others are a half, a quarter, an eighth and a sixteenth of it. Two quavers beamed together fill one pizza. A rest is the same share left empty: shaded with lines.

semibreve	whole note	4 pizzas · 4 beats	
semibreve rest	silence	4 empty pizzas · 4 beats	
minim	half note	2 pizzas · 2 beats	
minim rest	silence	2 empty pizzas · 2 beats	
crotchet	quarter note	1 pizza · 1 beat	
crotchet rest	silence	1 empty pizza · 1 beat	
quaver	eighth note	half a pizza · ½ beat	
quaver rest	silence	half an empty pizza · ½ beat	
semiquaver	sixteenth note	a quarter of a pizza · ¼ beat	
semiquaver rest	silence	a quarter of an empty pizza · ¼ beat	

Head, stem, flag

The **head** sits on a line or in a space. A **stem** goes up on the right of the head or down on the left. **Flags** on the stem halve the value each time; groups of quavers are joined by **beams** instead of flags.

Which rest is which?

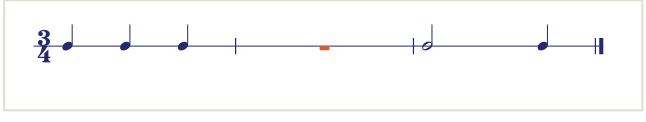


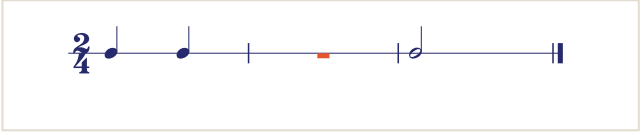
The semibreve rest **hangs** below the fourth line; the minim rest **sits** on the middle line.

THE SEMIBREVE REST'S SPECIAL POWER

Most rests last a fixed number of beats. The **semibreve rest** can also mean **a whole bar of silence** in any time signature: 4 beats in 4/4, but 3 in 3/4 and 2 in 2/4. It is then centred in the bar. No other rest can do this.

One more rule: keep the middle of a 4/4 bar visible. Silence on beats 2–3 is two crotchet rests, never one minim rest.

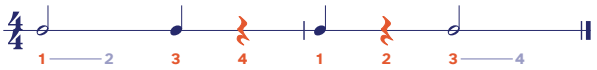




PART II · TIME · 2 · CONTINUED

EXAMPLE

How long is each sound, and each silence?

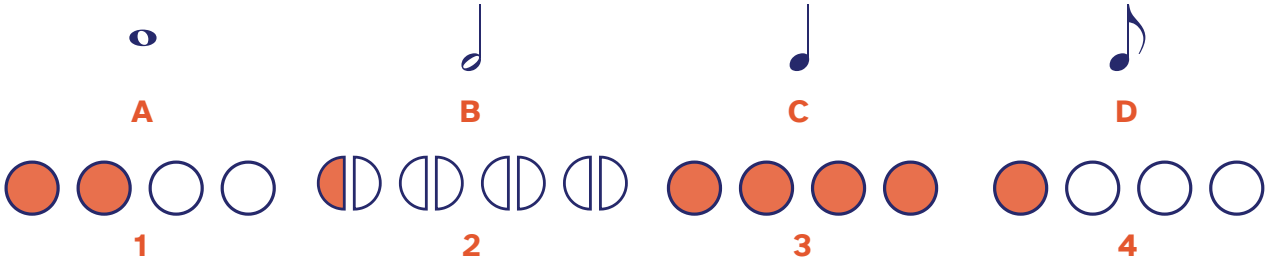


1. **Name each shape:** minim, crotchet, crotchet rest.
2. **Count its beats:** 2, 1 and 1.
3. **Count the rests too:** the red rests last as long as their notes.

Every bar adds up to **4 beats**, silences included.

GAME

Pizza match

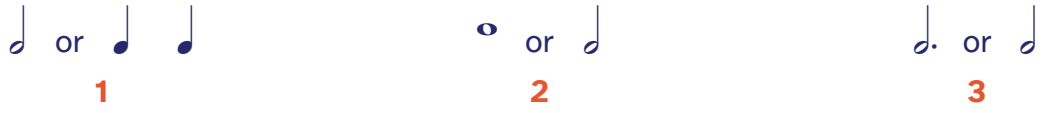


Match each note (A to D) with its pizzas (1 to 4). One pizza is one beat.

Answer: A 3, B 1, C 4, D 2.

GAME

Which lasts longer?



Circle the longer side in each pair. One pair is a trick: both sides are the same length.

Answer: 1 the same (2 beats each), 2 the semibreve (4 beats), 3 the dotted minim (3 beats).

TRY IT · 10 SECONDS



Clap once and hold for 2: a minim.

Now clap four crotchets (1 2 3 4), then one semibreve (clap, and hold all four beats). Say the counts out loud.

SEE ALSO
Pulse and the bar p. 34 · How long is a note? p. 39

PRACTISE IT
Workbook Level 0 · Note values · Online: Library › Notes

PART II · TIME · 3
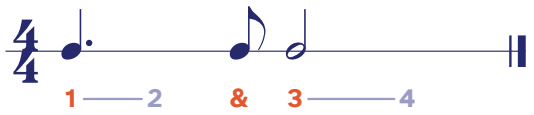
How long is a note?

Need a note longer than its shape allows? Two signs stretch it: the **dot**, which adds half, and the **tie**, which glues two notes into one sound.

THE SECRET
A dot adds half. A tie glues two notes into one.

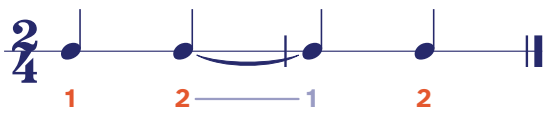
The dot

A dot adds **half the note's value**: a dotted minim lasts 2 + 1 = 3 beats; a dotted crotchet 1 + ½ = 1½. A **double dot** adds half, then a quarter: a double-dotted minim lasts 2 + 1 + ½ = 3½.



The tie

A curve joining two notes **of the same pitch** makes one longer sound: play the first, hold through the second. Ties can cross a bar line, where a dot cannot.
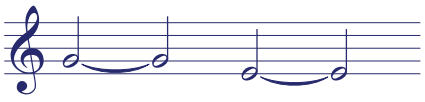


The dot, in pizzas

The dot is a helper that adds **half of whatever its note is worth**. Dark slices: the note. Lighter: what the dot adds. Lightest: a second dot, half of the first dot.

dotted minim		2 + **1** (half of 2) = **3 beats**
dotted crotchet		1 + **½** (half of 1) = **1½ beats**
dotted quaver		½ + **¼** (half of ½) = **¾ beat**
double-dotted minim		2 + **1** + ½ = **3½ beats**

Tie or slur?



Tie: the same pitch. Play once and hold: G for 4 beats, then E for 4.

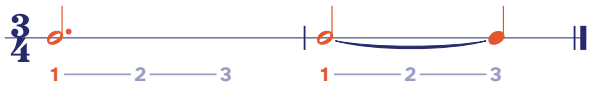


Slur: different pitches. Play every note, smoothly joined (legato).

PART II · TIME · 3 · CONTINUED

EXAMPLE 1

Dot or tie: which is longer?
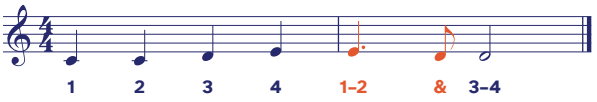


1. **The dotted minim:** 2 + half of 2 = 3 beats.
2. **The tied notes:** a minim tied to a crotchet, 2 + 1 = 3 beats.
3. **Play both:** one sound, held for three beats.

They are **the same length**. A tie can also cross a bar line, which a dot cannot.

EXAMPLE 2

A dotted crotchet you already know: Ode to Joy



1. **Bars 3–4 of Beethoven's Ode to Joy.** Sing it: "C C D E, E — D D".
2. **The red E is a dotted crotchet:** 1 + ½ = 1½ beats, held through "1 & 2".
3. **The quaver D** falls on the "&" of beat 2; the last D takes beats 3 and 4.

Long–short, then long: the dot is why the tune **leans** before it lands.

EXAMPLE 3

Which is longer: a dotted minim or a minim tied to a quaver?



1. **Dotted minim:** 2 + 1 = 3 pizzas.
2. **Minim tied to a quaver:** 2 + ½ = 2½ pizzas.
3. **Compare:** 3 is half a beat more than 2½.

The **dotted minim** is longer, by half a beat.

THINK OF IT A tie couples two carriages: the passenger stays seated across the join, even over a bar line. A dot cannot cross the bar line.

TRAP

- ✕ Putting the dot on the wrong note: a dot adds half of **its own** note.
- ✕ Tying two different pitches: that curve is a slur, not a tie.

SEE ALSO

Time signatures p. 41 · Rhythm p. 45

PRACTISE IT

Workbook Level 0 · Note values · Online: Library › Notes and rests

PART II · TIME · 4

Time signatures

Two numbers, like a train ticket: the **top number** says how many seats (counts) each bar has; the **bottom number** says which note fills one seat.

THE SECRET

Top number: how many beats. Bottom number: which note gets one beat.

The bottom number

BOTTOM	NOTE	ONE COUNT IS A	FIRST MET
2		minim	Level 3
4		crotchet	Level 0
8		quaver	Level 1
16		semiquaver	*Extra*

Simple time: each beat divides in two

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
2/4	2 crotchets	Level 0
3/4	3 crotchets	Level 0
4/4 (C)	4 crotchets (the C sign: common time)	Level 0
2/2 (¢)	2 minims (the cut C: cut common time)	Level 3
3/2	3 minims	Level 3
4/2	4 minims	*Extra*
3/8	3 quavers	Level 3

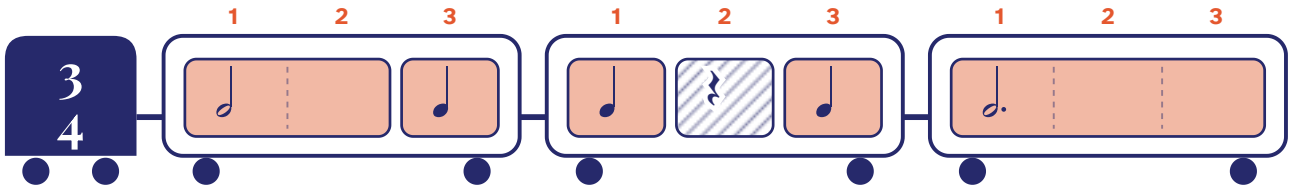
Changing time signature

A new time signature can appear in the middle of a piece, right after a bar line; it holds until the next one. A time signature is written only once, not on every line.

PART II · TIME · 4 · CONTINUED

Think of a train

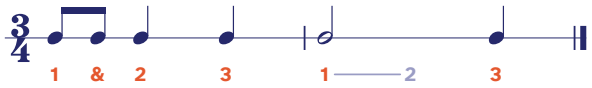
In 3/4 every carriage (bar) has exactly **3 seats**; each seat is one crotchet beat. Notes book seats: a minim takes 2, a crotchet 1. A rest books a seat and leaves it **empty** (shaded). Every seat in every carriage must be booked.



minim + crotchet · crotchet, crotchet rest, crotchet · dotted minim: 3 seats each

EXAMPLE

Read a time signature
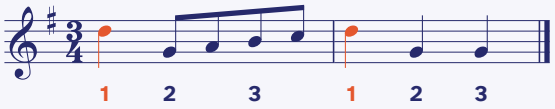


1. **Top number 3:** three counts in each bar.
2. **Bottom number 4:** a crotchet is one count.
3. **Check a bar:** two quavers + crotchet + crotchet = 3.

3/4: **three crotchet beats** in every bar.

YOU KNOW THIS

Minuet in G (from the Notebook for Anna Magdalena Bach)



Every bar holds **three crotchet beats**: 1 2 3, 1 2 3, the step of the dance. That is 3/4.

Try it. Count 1 2 3 aloud while someone plays it.

TRAP

- × Reading the bottom number as a count: it names the note that gets one beat.
- × Reading 6/8 as six beats: at most speeds it moves in two.
- × Mixing up 3/4 and 6/8: both hold six quavers, grouped differently (p. 43).

SEE ALSO

Pulse and the bar p. 34 · Compound time p. 43

PRACTISE IT

Workbook Level 0 · Time signatures · Online: Library › Time signatures

PART II · TIME · 5

Compound and irregular time

Compound time skips: each dotted beat splits into **three**. Irregular time keeps you guessing, with **five or seven** counts in uneven groups. (9/4, 12/4 and /16 signatures are rare.)

THE SECRET

Dotted beats split in three: 6/8 is two big beats.

3/4 or 6/8?



3/4: six quavers in three twos



6/8: six quavers in two threes

SIMPLE OR COMPOUND?
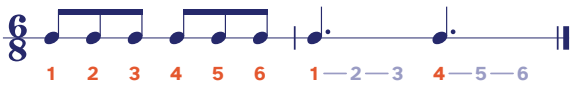
In **simple time** each beat divides into **two**. In **compound time** each beat is a dotted note and divides into **three**.

Quick test: a top number of 6, 9 or 12 means compound. Divide it by 3 to find the beats: 6/8 has 2, 9/8 has 3, 12/8 has 4.

Compound time

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
6/8	2 dotted crotchets	Level 1
9/8	3 dotted crotchets	Level 3
12/8	4 dotted crotchets	Level 3
6/4	2 dotted minims	Level 4

Counting 6/8



Slowly, count all six quavers. At speed, feel two beats, on 1 and 4: “1 & a, 2 & a”.

Irregular time

UNEVEN GROUPS IN 4/4

Many dances keep 4/4 but group the eight quavers 3 + 3 + 2: “1 2 3, 1 2 3, 1 2”.



PART II · TIME · 5 · CONTINUED

TIME SIGNATURE	EACH BAR HOLDS	FIRST MET
5/4	5 crotchets: 3 + 2 or 2 + 3	Level 5
7/4	7 crotchets: 4 + 3 or 3 + 4	*Extra*
5/8	5 quavers: 3 + 2 or 2 + 3	*Extra*
7/8	7 quavers: 2 + 2 + 3 or 3 + 2 + 2	*Extra*

EXAMPLE 1

Count 6/8 in two

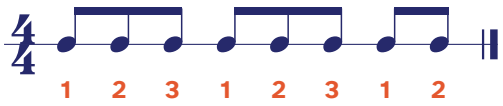


1. **Slowly, count all six:** 1 2 3 4 5 6.
2. **Lean on 1 and 4**, the red notes.
3. **Faster, count only those:** two beats, each a dotted crotchet.

6/8 has **two beats** in a bar, each split into three.

EXAMPLE 2

Hear 3 + 3 + 2 inside 4/4
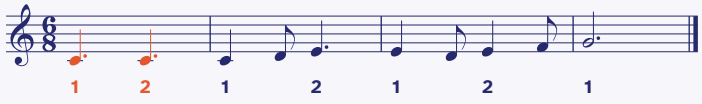


1. **Count eight quavers** in a bar of 4/4.
2. **Group them** 3 + 3 + 2, accenting 1, 4 and 7.
3. **Keep the beat** in your foot: it stays in four.

The bar is still **4/4**; only the accents are uneven.

YOU KNOW THIS

Row, Row, Row Your Boat



Two big beats in every bar, each a **dotted crotchet**: “Row, row”. Inside each beat, “gent-ly” splits it into three. That is 6/8.

Try it. Sway twice per bar while you sing it.

THINK OF IT Like a skipping rope: two big swings, each with three little steps inside.

SEE ALSO

Time signatures p. 41 · Rhythm: groups and upbeats p. 45

PRACTISE IT

Workbook Levels 1–4 · Compound time · Online: Rhythm clock

PART II · TIME · 6

Rhythm: groups and upbeats
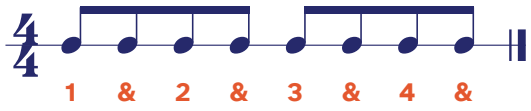
Say your own name: long and short sounds, that's **rhythm**. The pulse marches steadily on; the rhythm dances around it. How the notes are grouped on the page shows where the beats fall.

THE SECRET
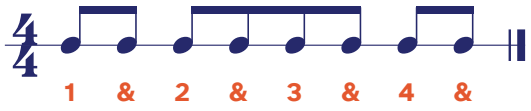
Beams group notes into beats; an upbeat borrows from the last bar.

Grouping with beams

Beams join the notes of **one beat**, so the beats stay visible. In 4/4 a beam may cover beats 1–2 or 3–4, but never cross the middle of the bar (except in a deliberate grouping such as 3 + 3 + 2). In compound time, quavers are beamed in threes.



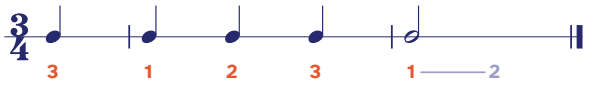
✓ The middle of the bar can be seen.



× A beam across beats 2–3 hides it.

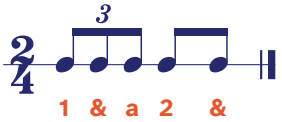
The upbeat (anacrusis)

A piece can begin **before** the first full bar, with one or more notes called an upbeat or anacrusis. The last bar is then short by the same amount, so the two together make one full bar. Count the missing beats in silence before you start.
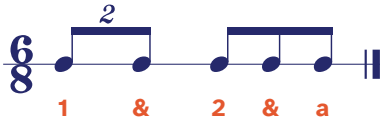


Count "1 2" silently, then play on 3.

Triplets and duplets



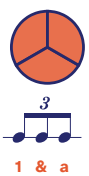
Triplet: three exactly equal notes in the time of two. First met: Level 2.



Duplet: two notes in the time of three, in compound time. First met: Level 2.



Two quavers: one beat cut in 2



A **triplet**: one beat cut in 3



Four semiquavers: one beat cut in 4

PART II · TIME · 6 · CONTINUED

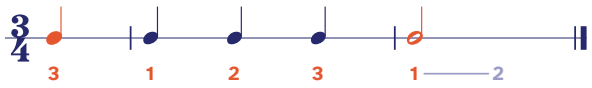
Counting smaller values



Crotchet, two quavers, four semiquavers and a triplet (“1 & a”): one beat each.

EXAMPLE 1

Balance the upbeat
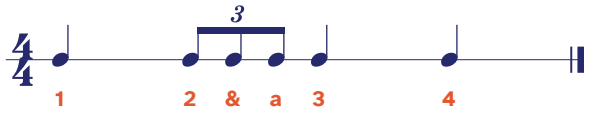


1. **The upbeat:** one beat (count 3) before the first full bar.
2. **The last bar:** only two beats.
3. **Add them:** 1 + 2 = 3, one whole bar of 3/4.

The upbeat and the last bar **make one full bar** together.

EXAMPLE 2

Fit a triplet into a bar



1. **Count the bar:** four crotchet beats, “1 2 3 4”.
2. **Find the 3:** on beat 2, three quavers share one beat.
3. **Say it:** “1, 2 & a, 3, 4”, keeping beats 3 and 4 steady.

The triplet squeezes three notes into **beat 2**; the beat itself never moves.

YOU KNOW THIS

Happy Birthday



“Hap-py” comes **before** the first strong beat: an upbeat. The strong beat lands on “birth”.

Try it. Sing it and clap only on “birth”, “day” and “you”: you feel where bar 1 starts.

THINK OF IT An upbeat is the breath before you speak: “and-ONE”. You lift on the upbeat so the strong beat can land.

SEE ALSO
Notes and rests p. 36 · Rhythm: off the beat p. 47

PRACTISE IT
Workbook Levels ½–4 · Rhythm · Online: Rhythm clock

PART II · TIME · 7

Rhythm: off the beat

Now the fun part: rhythm that teases the beat. It only works if the beat underneath stays rock-steady.

THE SECRET

Keep the beat rock-steady and let the rhythm tease it.

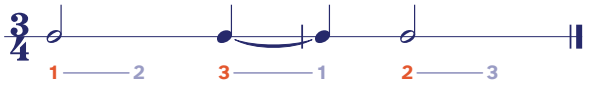
Syncopation



The crotchet starts on “&” and is held through beat 2.

Level 4. Feel beat 2 even though nothing is played on it.

Hemiola



Two bars of 3/4 felt as three minims.

Level 4. Common at cadences in Baroque dances.

Swing Level ½

Common from the first pieces: jazz, blues and many easy modern pieces are marked *swing*. Pairs of quavers are written evenly but played **long–short**, like the first and last notes of a triplet.

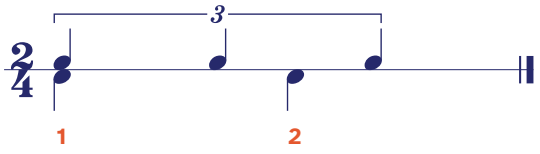


Written



Played

Polyrhythm Level 7



Three against two: a crotchet triplet above, two crotchets below, starting together.

Rubato Extra

Time is stretched and then given back, for expression: the music leans into a phrase and catches up after it. The pulse always returns. Marked *rubato*, or simply part of the style in much Romantic music.

PART II · TIME · 7 · CONTINUED

EXAMPLE 1

Count a syncopation
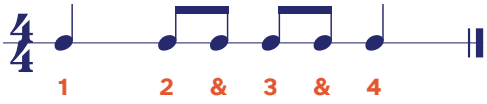


1. **Tap the beats:** 1, 2.
2. **Find the red crotchet:** it starts on the "&", between the beats.
3. **Hold it through beat 2**, then play bar 2 on the beats.

The accent falls **off the beat**: that is syncopation.

EXAMPLE 2

Swing this bar

Swing
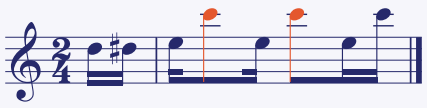


1. **Find the quaver pairs:** beats 2 and 3.
2. **Play each pair long–short:** "doo-ba, doo-ba".
3. **Leave the crotchets alone:** beats 1 and 4 stay as written.

Only the **pairs** swing; the beat underneath stays even.

YOU KNOW THIS

The Entertainer (Scott Joplin)



The high **C** keeps landing just after the beat, not on it: syncopation, the bounce of ragtime.

Try it. Tap the beat with your foot and hum the tune: hear the C slip between the taps.

THINK OF IT Syncopation is a skip in your walk: a step that lands between the beats. Swing is walking with a spring in your step, long-short, long-short. Rubato is borrowing time like pocket money: whatever you spend, you pay back.

TRAP

- × Rushing the short notes after a long one: keep counting inside the long note.
- × Playing a triplet unevenly: three equal notes, not two quick and one long.
- × Letting syncopation drag the pulse: tap the beat with your foot.

SEE ALSO

Rhythm: groups and upbeats p. 45 · Words for how to play: Expression (later)

PRACTISE IT

Workbook Levels ½–4 · Rhythm · Online: Rhythm clock

PART II · TIME · TIPS

When the rhythm slips

A rhythm that slips is nearly always a beat that was not counted. Find which beat, count it out loud, and the rest of the bar usually falls into place.

TIP

What you notice, what to check

WHAT YOU NOTICE	CHECK	TURN TO
It speeds up or slows down	**the pulse**: count one bar aloud before you start	p. 34
A rest gets skipped	**the rest**: it is counted just like a note	p. 36
Long notes come out short	**the hold**: count every beat of a minim, dot or tie	p. 39
The bar feels too long or too short	**the time signature**: how many counts, which note is one	p. 41
An off-beat note lands on the beat	**the "&"**: keep the beat in your foot and count it	p. 47

Four ways to steady a rhythm

1. **Count out loud**
   Say the counts while you play, not in your head: a slip you can hear is a slip you can fix.

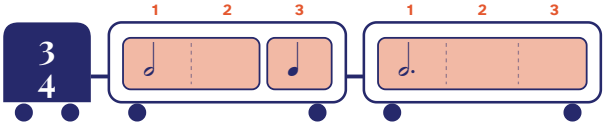
2. **Tap, then play**
   Tap the rhythm on the lid and say the counts; play only when the tapping is steady.

3. **Walk the beat**
   Step the pulse while you clap the rhythm, so the beat lives in your feet.

4. **Split the beat**
   For dots and quavers, count "1 & 2 &": the extra "&" shows exactly where each note goes.

EXAMPLE

Count the long notes



1. **Say 1 2 3** once through before you play, to feel the beat.
2. **The minim books seats 1 and 2:** keep saying "2" while it sounds.
3. **The dotted minim books all three:** let go on the next "1", not before.

Long notes are where counting slips: **say every seat** they book.

THINK OF IT Like a dance partner: the beat leads and the rhythm follows. Lose the beat and every step after it goes wrong.